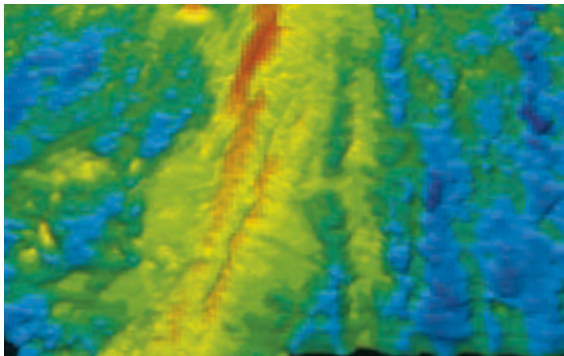




▲ BILLOWING LAVA

Basalt lava squirts up through fissures in the spreading rift at temperatures of about 1,800°F (1,000°C), but then hits deep-ocean water that is close to freezing point. The lava hardens instantly on the outside, but then bursts out through the hardened shell to form a series of rounded lobes. When these finally become solid they look like black cushions, so they are called pillow lavas. In a few places, where the Earth's crust has been uplifted, these can be seen on land.



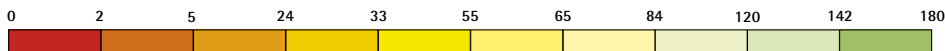
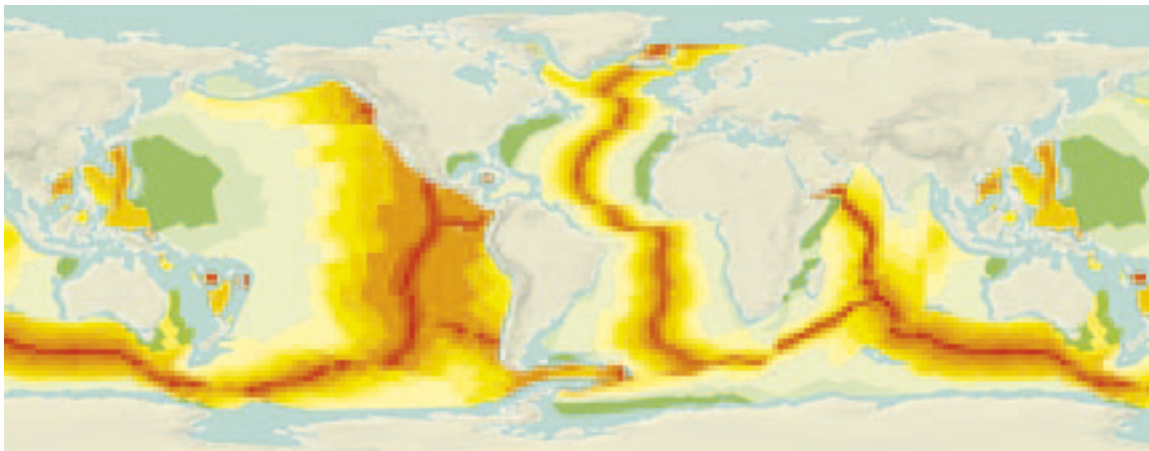
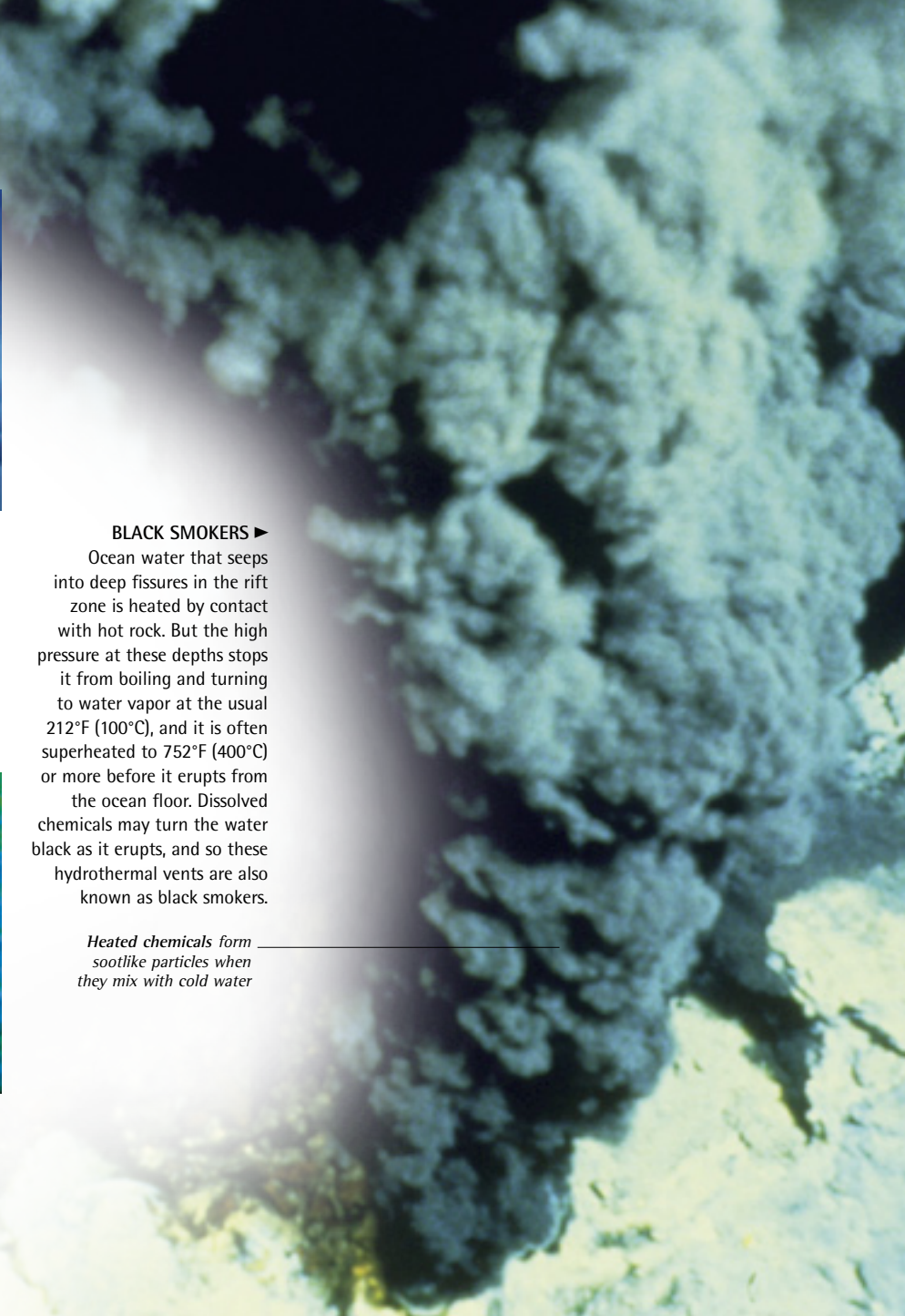
▲ THE EAST PACIFIC RISE

The Mid-Atlantic Ridge is spreading at about 2 in (4 cm) a year, but other midocean ridges are spreading faster. The East Pacific Rise to the south of Mexico, seen here in a sonar image, is spreading at 9 in (22 cm) a year. Yet, although it is more active, its submarine mountains are not so high, because the hotter, softer rock beneath the ridge cannot support big, heavy peaks. The rift is peppered with underwater volcanoes and clusters of hydrothermal vents.

BLACK SMOKERS ►

Ocean water that seeps into deep fissures in the rift zone is heated by contact with hot rock. But the high pressure at these depths stops it from boiling and turning to water vapor at the usual 212°F (100°C), and it is often superheated to 752°F (400°C) or more before it erupts from the ocean floor. Dissolved chemicals may turn the water black as it erupts, and so these hydrothermal vents are also known as black smokers.

Heated chemicals form sootlike particles when they mix with cold water



NOT DATED

◀ OCEAN FLOOR AGE

Ocean floor is formed at midocean ridges, and gradually moves away from them on both sides as more rock erupts from the spreading rifts. So the rocks near the ridges are the youngest, and those farther away get progressively older. But no part of the ocean floor is more than 180 million years old, because any oceanic rocks formed before this time have been drawn back into the mantle and recycled. By comparison, the oldest continental rocks are 3,800 million years old. The map and key show the ocean floor age in millions of years.